MIL-R-24037/5A(EC)
21 December 1984
SUPERSEDING
MIL-R-24037/5(SHIPS)
29 May 1964

MILITARY SPECIFICATION

RADIACMETER, IM-143( )/PD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the dosimeter described herein shall consist of this specification and the latest issue of MIL-R-24037.

1. SCOPE

1.1 Scope. This specification covers an ion-chamber quartz-fiber dosimeter (radiacmeter), capable of measuring and indicating doses of gamma or X-radiation over the range of 0 Roentgens (R) to 600 R.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification. Unless otherwise specified, the following specification of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-24037	Radiacmeters, Quartz-Fiber Self-Indicating, General Specification

(Copies of the specification required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3 1 The dosimeter shall be in accordance with MIL-R-24037 and as specified in 3.2 through 3.8.

3.2 Scale. The scale shall have 30 equal divisions, each division representing 20 R. There shall be seven major marking lines. 0, 100, 200, 300, 400, 500, and 600. The major marking lines shall be marked with Arabic numerals.

3.3 Range. The range of the dosimeter shall be 0 R to 600 R.

* 3.4 Leakage. In the absence of radiation, the electrical leakage rate shall not exceed 6.0 R per day for any relative humidity to 100 percent and for the temperature range from -65° Fahrenheit (F) to 100°F. At 131°F, the leakage rate shall not exceed 9.0 R and shall be scaled linearly from 6.0 R to 9.0 R in the temperature range from 100°F to 131°F.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6665

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3.5 Leakage after heavy irradiation. The leakage over the 24-hour period beginning 8 hours after an irradiation of 5000 Rs at a rate of 500,000 Rs per hour of X-rays from an unfiltered 250-kilovolts-peak machine shall not exceed 5 percent of full scale.

3.6 Accuracy. The dosimeter shall measure true gamma dose from a cobalt 60, radium, or cesium 137 source within ±10 percent or ±15 R, whichever is larger. This accuracy requirement shall apply for any source which has been:

a. Calibrated directly by the National Bureau of Standards (NBS)
b. Calibrated by the contractor using a Victoreen R meter, or equivalent, approved by the command or agency concerned which in turn have been calibrated by the NBS
c. Calibrated by a Government activity using a Victoreen R meter, or equivalent, approved by the command or agency concerned which have been calibrated by the NBS

Specifically included in this spread are the usual imprecisions involved in correlating source strengths with the standard NBS sources.

3.7 Energy dependence. Except at the energies represented by cesium 137, and cobalt 60, where the requirements of 3.6 shall apply, the dosimeter shall indicate true gamma dose within ±15 percent or 15 R, whichever is larger, for X- or gamma radiation of any energy from 80 kilo-electronvolt (keV) to the energy of cobalt 60 gammas. Over the range 80 keV to 50 keV, the dosimeter response shall decrease smoothly, at 50 keV and at all X- and gamma radiation energies below 50 keV the response shall be not greater than 50 percent of true dose. As used here, 80 keV and 50 keV refer to effective energies of X-ray beams from a constant voltage X-ray machine with appropriate tube voltage and filtration in accordance with normal NBS procedures.

3.8 Ion chamber saturation and rate dependence. The dosimeter, when subjected to dose rates up to 600,000 R per hour, shall not vary from their normal calibration accuracy by more than ±6.0 R.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance provisions shall be in accordance with MIL-R-24037, and as specified in 4.2 through 4.4.

* 4.2 Leakage. The dosimeter shall be fully charged and tested for a period of 5 days at 100°F. Again, the dosimeter shall be fully charged and tested for a period of 5 days at 131°F. The leakage shall not exceed 30 R and 45 R respectively during this test. The dosimeters may be stored under charge for some time prior to this test if desired. The fibers shall be charged positively in this and all other tests.

4.3 Calibration at mid point only. A source, calibrated in accordance with 3.6 shall be used. Temperature may be normal room temperature. A dose of 300 R shall be delivered to the dosimeter. The reading shall be not greater than 330 R nor less than 240 R.

4.4 Calibration over the entire range. The instrument shall be tested as specified in 4.3 at five points: 60 R, 180 R, 300 R, 420 R, and 540 R Readings shall be as specified in 3.6.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preparation for delivery shall be as specified in MIL-R-24037.

6. NOTES

6.1 Notes. The notes specified in MIL-R-24037 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
NAVY-EC

(Project No. 6665-N475)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-24037/5A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*. ____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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